ELECTRICAL AND ELECTRONIC MEASUREMENTS

INTRODUCTION

Electrical and electronic measurements of materials are among the most powerful techniques available for materials characterization. These measurements can reveal information that completely characterizes the electrical transport properties of a material. Furthermore, modern instrumentation makes precision measurements and computer interfacing straightforward at reasonable cost.

Conductivity measurements yield information on the conductivity (resistivity) of materials and indirectly the mobility of current carriers. Resistivity is one of the most sensitive measures of the electrical transport in materials and can vary from 10^{12} ohm-cm for the best insulators to 10^{-6} ohm-cm in pure normal metals. Measurements include both DC and AC techniques. The Hall effect is a versatile technique, yielding such information as carrier concentrations and mobility, and indirectly, estimates of scattering mechanisms, all of which affect the conductivity as a function of temperature. Careful experimental technique can also yield information about the ionization energy of dopants contributing to the electrical transport. These measurements, used in conjunction with solid-state spectroscopies, can yield information about the band structure of semiconductors.

The capacitance-voltage (C-V) measurement is the standard method of "profiling" free carrier concentrations as a function of position below the surface. A C-V measurement is the most common of the carrier profiling techniques due to its simplicity and low cost. Deep level transient spectroscopy (DLTS) is a transient capacitance measurement, which permits identification of deep traps within the forbidden gap of a semiconductor. This highly effective and elegant measurement technique yields information regarding the energy levels of these defects, their capture and emission rates, and their concentration. Equipment for measuring both C-V and DLTS are often combined in a single apparatus. The use of C-V measurements, DLTS characterization, and the Hall effect yields a potent suite of characterization techniques that can completely describe the electronic transport properties of a material.

Semiconductor materials can be prepared with either electrons or holes as the majority current carriers. Most measurement techniques are useful in characterizing a single carrier type, or, at best, the net carrier type for mixed transport. It is often useful to gain insight into transport across a junction formed between materials at which the carrier type changes abruptly from holes (p-type material) to electrons (n-type material). Transport measurements at pn–junctions yield information on the transport of both holes and electrons and can determine transport mechanisms as a function of carrier (current) density.

The carrier lifetime is one of the most sensitive measurements available for detecting impurities in semiconductors. At its most sensitive, this materials-characterization technique is capable of detecting impurity concentrations as low as one impurity atom per 10^{12} host atoms. Techniques available for carrier lifetime measurements are varied and plentiful. Optical techniques rely upon excitation of the material from its equilibrium state, typically by using a short optical (laser) pulse, then carefully measuring the return to equilibrium. Because the optical excitation can be finely focused, these techniques have the added ability to spatially map a material.

Transport measurements, particularly those techniques that yield information on carrier concentration, mobility, and scattering mechanisms, are useful educational tools for undergraduate science and engineering students. Students can perform experiments that reinforce lectures in courses such as solid state physics, physical chemistry, materials science, and device physics. Furthermore, the use of such apparatus as high input-impedance electrometers, capacitance meters, phase-sensitive detectors, lasers for optical excitation, spectrometers, and photon counting equipment greatly enriches students' knowledge of the equipment and instrumentation used in the study of materials. Thus, experiments centered upon materials characterization prepares a student for graduate study or entrance into the work force. The student is equipped with both knowledge of electrical transport in materials and the sophisticated yet inexpensive equipment used in their measurement.

PETER A. BARNES

CONDUCTIVITY MEASUREMENT

INTRODUCTION

A material's conductivity, σ, (or the inverse property, resistivity, ρ, where $\rho = 1/\sigma$,) relates to its ability to conduct electricity. In metals, conduction of electricity is tantamount to conduction of electrons, which depends on charge density and on scattering of the electrons by the crystal lattice (phonons) or by lattice imperfections. In semiconductors, conductivity is determined by the number of available charge carriers (electrons or holes) and the carrier mobilities. Because of the different mechanisms for conductivity, its dependence on temperature also differs. Conductivity increases with increasing temperature for semiconductors (more carriers are generated) and it decreases with increasing temperature for metals (more scattering by the lattice). Conductivity also depends on

physical structure. In crystals, the crystal type and orientation affect conductivity because the electronic structure is intimately tied to the crystal structure. The size of the crystallites (grains) in polycrystalline materials is also important as it affects the scattering of carriers, and, at very small sizes may also affect electronic structure.

For a rectangular slab of homogeneous material with length L, width W, and thickness t (in m), the resistance, R (in ohms, Ω) along the length of the slab is related to the resistivity ρ in Ω-m by

$$R = \frac{\rho}{t}\frac{L}{W} \tag{1}$$

Units for conductivity are most often expressed in terms of siemens per meter (S/m), or by the equivalent terms 1/(Ω-m) and (less common) mhos/m. In the semiconductor field, the quantity ρ/t is termed sheet or surface resistance, R_s, and quoted in units of Ω per square. The number of "squares" of a thin-film conductor or resistor material is given by the ratio L/W.

The accurate determination of a material's conductivity can be critical for understanding material composition or device performance. The method used to determine conductivity depends on whether the material is a bulk sample or a thin film. This unit begins with a section on bulk measurements (see the discussion of Bulk Measurements). Two-point measurement using an inexpensive ohmmeter is perhaps the simplest approach, although its accuracy is limited and fairly large samples are required. A four-point bulk measurement is only slightly more complicated, but yields much better accuracy. Both of these methods are potentially destructive to the sample, depending upon how electrical contacts are made. These measurements are described along with advice on how to make the most accurate measurements in different situations.

In the second section (see discussion of Surface Methods), two thin-film approaches are presented which are most frequently used in semiconductor characterization: four-point probing and the Van der Pauw method. These methods require more expensive test equipment but are very accurate. Four-point probing requires little sample preparation, although it can cause damage to the surface. The Van der Pauw approach generally requires the creation of a test pattern on the surface but can take up less test space on a semiconductor than four-point probing.

In the third section (see discussion of Non-Contact Methods) measurement approaches are presented that offer the possibility of measurements without damaging the material. These approaches are not as well developed as the contact approaches described in the first two sections. Specialized equipment is required, and accuracy can be rather limited. Extracting conductivity values often depends upon qualitative comparison to standard data. The most widely used non-contact approach is the eddy current method, which can be used for metals and semiconductors, and which is implemented in a variety of ways. Also described in this section is the relaxation method.

The fourth section (see discussion of Microwave Techniques) briefly describes the most common microwave techniques that can be employed to determine conductivity information at high frequencies. Expensive test equipment is required for these approaches, along with specialized test apparatus to probe or hold the samples. The conductivity is not directly measured, but is typically extracted from loss tangent determination.

Some general characteristics for the methods described in these four sections are listed in Table 1.

Electrical conductivity (resistivity) is a fundamental materials property and as such finds its value incorporated in many complex and sophisticated treatments of materials

Table 1. General Characteristics for Conductivity Measurement Methods

Method	Favoredmaterial Type	Favored Material Form	Estimated Measurement Range	Estimated Frequency Range
Two-point measurement	High-resistance metals	Solid bar	10^2–10^9Ω	DC AC (<300 Hz)
Four-point measurement	Metals	Solid bar	10^{-7}–10^4 Ω	DC AC (<300 Hz)
Four-point probe	Semiconductor surface, thin metallic films	Planar solid	10^{-3}–10^5 Ω-cm	DC
Van der Pauw	Semiconductor surface, thin metallic films	Planar solid	10^{-3}–10^5 Ω-cm	DC
Eddy current	Metals, semiconductors, Insulators	Solid wafer, cylinder, containerized liquid	10^{-10}–10^{10} Ω-cm	10–10^6 Hz
Relaxation	Metals, semiconductors	Solid wafer, containerized liquid	10^{-8}–1 Ω-cm	10^{-1}–10^7 Hz
Coaxial probe	Semiconductors, insulators	Flat surfaced solid, liquid	10^{-3}–10^5 Ω-cm	2×10^8 to 2×10^{10} Hz
Transmission line	Semiconductors, insulators	Precision-machined sample, containerized liquid	10–10^6 Ω-cm	10^8–10^{10} Hz
Free-space radiation	Metals, moderate to high loss materials	Large area, flat solid sample	10–10^5 Ω-cm	10^9–10^{11} Hz
Cavity resonator	Metals, moderate to high loss materials	Machined solid sample	10–10^8 Ω-cm	5×10^8 to 10^{11} Hz

behavior. Whereas this value is often "plugged-in" from a table of known properties, it may also be deduced from the governing relations if other pertinent factors are known. This unit does not attempt to describe the universe of indirect measurement possibilities in regard to conductivity and resistivity.

BULK MEASUREMENTS

Principles of the Method

The conductivity or resistivity of a bulk sample is based on accurate measurement of both resistance and the sample dimensions. The resistance is the ratio of the voltage measured across the sample to the current driven through the sample or of the voltage applied across the sample to the measured current.

For a homogeneous bar of length, L, and uniform cross-section, A, the resistance, R, is related to the resistivity, ρ, by

$$R = \rho L/A \tag{2}$$

In Figure 1A, the bar is connected in a "two-point" arrangement, so-called since the measurement apparatus is connected to the bar at the two end-points. The measurement apparatus is represented by an ideal current source in parallel with a high-impedance voltmeter. The apparatus can also be realized using an ideal voltage source in series with a low-impedance ammeter, as shown in Figure 1B. A more realistic view of a two-point measurement using an ohmmeter is shown in Figure 2. A voltage source and a variable range resistor (R_r) supply the current, where R_r is adjusted to provide a convenient voltage across the voltmeter. Typical values of R_r range from 100 to 10,000 Ω. R_c represents series resistance in the cable and the wire-to-sample contact resistance. The resistance in the bar is calculated as

$$R = \frac{(R_r V_m/V_s)}{[1 - (V_m/V_s)]} - 2R_c \tag{3}$$

A long bar of resistive material is desirable to minimize the effect of extra resistance in the measurement system or inaccurate length measurement.

A more accurate method, especially for low-resistance materials, is provided by the "four-point" arrangement, a simple schematic of which is shown in Figure 3. The resistance now is a function of the length between the voltage probes, L', or

$$R = \rho L'/A \tag{4}$$

The current through the high-impedance voltmeter is very small and minimizes the effect of cable and contact resistance. The four-point approach, when used with a micro-ohmmeter, is capable of measuring down to 10^{-7} Ω.

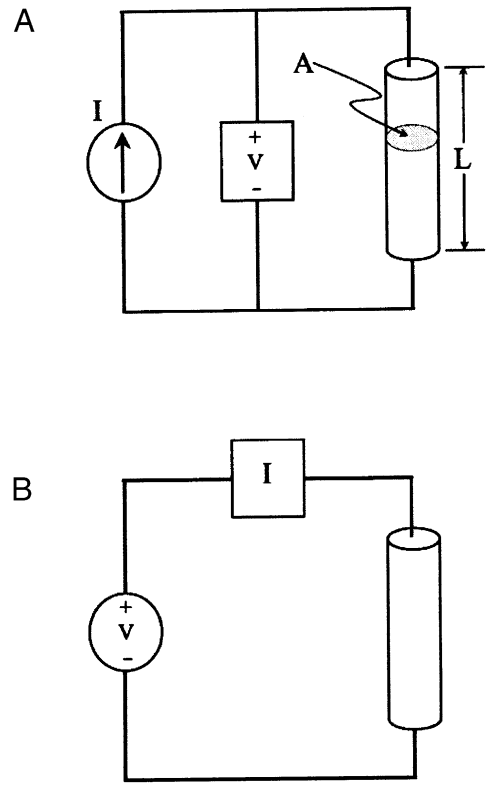


Figure 1. Two-point measurement of a resistive bar of length, L, and cross-sectional area, A, using (**A**) an ideal voltmeter in parallel with an ideal current source, and (**B**) an ideal ammeter in series with an ideal voltage source.

Practical Aspects of the Method

The two-point approach is most accurate for high-resistance measurements where the usually small R_c term (generally less than 2 Ω) can be ignored. These measurements are often made using ohmmeters contained within multimeters capable of measuring voltage, current, and resistance. The ohmmeter within the multimeter performs like the measurement circuits shown in Figure 1, and is realized using a variety of circuit implementations [see, e.g., Witte (1993) or Coombs (1995)]. Inexpensive multimeters (<$50) can measure resistance between 0.1 Ω and

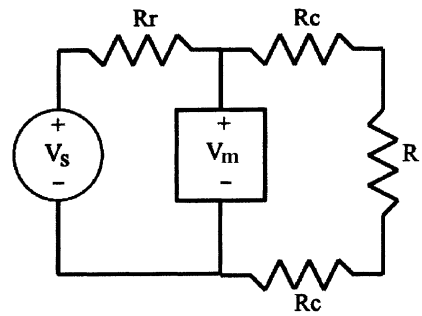


Figure 2. Two-point ohmmeter measurement circuit, which includes contact and cable resistance, R_c.

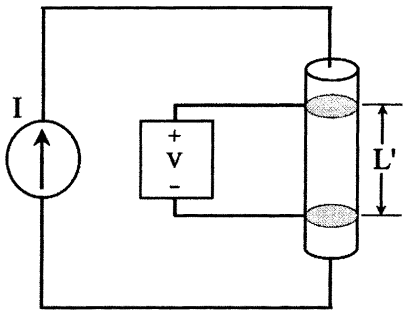


Figure 3. Four-point measurement circuit drives a known current through the sample, and a high-impedance voltmeter measures voltage drop along a designated length, L'.

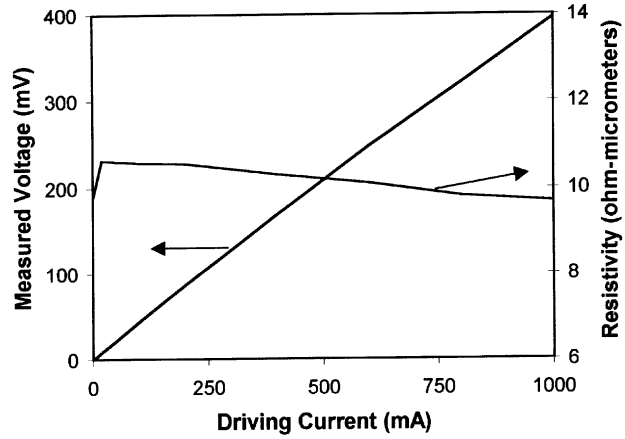


Figure 4. Measured voltage across a 2.5-cm length section of graphite rod (0.0889 cm diameter) versus the driving current, along with the extracted resistivity.

10 kΩ. More expensive meters (~$1000) can measure 100 μΩ to 100 MΩ and are provided with an extra pair of terminals for performing four-point measurements, described below. Dedicated ohmmeters can also be purchased for special measurements. For instance, high-resistance meters are capable of measurements up to 10^9 Ω. These meters require large, precision voltage sources (in the vicinity of 1000 V) along with very low–current measurement capability, perhaps as low as 10^{-15} A. Also, since ohmmeters have internal capacitance, C, the large time constant, $\tau = RC$, for a high-resistance measurement can be excessive and it may be extremely difficult to make a steady-state measurement.

At low resistance, the cable and contact resistance can seriously degrade measurement accuracy. Some ohmmeters have a "zero-adjust" feature, where the test leads are shorted together and the meter is adjusted to zero. Such an adjustment compensates for cable resistance and for measurement drift internal to the ohmmeter. In a related approach, the user can measure a short circuit, perhaps employing the same electrical contact strategy as for the actual test device, in order to determine R_c. Another approach to account for R_c is to make measurements on a pair of devices having the same cross-section but different lengths. The difference in resistance between the two measurements should result in the resistance due only to the difference in length of the bar.

In the four-point approach, it is im portant to keep the voltage contacts very thin so an accurate measurement of L' can be made. Voltage probes should be well away from the current probes to avoid "spreading resistance" interference. This spreading resistance is a consequence of the current spreading radially from the probe tip into the sample, with a considerable amount of resistance encountered near the tip. As a rule of thumb, current and voltage probes should be separated by a distance of at least 1.5× the thickness of the measured sample (ASTM B193, 1987).

As an illustration of bulk measurement, a 6-cm-long graphite rod of diameter 0.0889 cm was measured at room temperature using a two-point ohmmeter. Measurement of 1.9 Ω on a 5.8-cm length (distance between the two mechanical clips) yielded a resistivity of 20 Ω-μm. No attempt was made to remove cable and contact resistance from this measurement. Next, a four-point measurement was made using a driving current ranging from 1 mA up to 1 A that was connected to each end of the rod, and a voltmeter connected at points 2.5 cm apart near the center of the rod. The measured voltage as a function of driving current is shown in Figure 4. The resistivity from this four-point approach is ~10 Ω-μm. This value is about half of that measured using the two-point approach, indicating the presence of considerable contact and/or cable resistance that is avoided by the four-point approach. There is slight curvature on the voltage plot, resulting in a measured drop in resistivity at higher currents. This can be explained by graphite's decrease in resistance as a function of temperature, since the rod heats up as more current is driven through it.

Method Automation

A number of companies make digital multimeters, current sources, and voltage meters with GPIB (general purpose interface bus) connections suitable for test control and automatic data recording. The software package LabView (National Instruments Corp.) is routinely employed to handle such measurements.

Sample Preparation

Making an accurate resistance measurement requires good electrical contacts to the sample. A minimum spacing between voltage and current contacts of 1.5× the cross-sectional perimeter of the sample is recommended (ASTM B193, 1987), and care should be taken to prevent any alternative conductive path (for instance, a solder bridge between a current and a voltage probe). Making good electrical contacts often consists of mechanical contact (i.e., alligator clips or probing with a sharp probe tip) or soldering a wire to a clean surface on the sample. For troublesome surfaces, silver paint or indium (a soft metal) can provide a contacting layer.

Problems

Even with good contacts and a good four-point measurement setup, other factors should be considered. For instance, appreciable power dissipation in the resistive material can result in heating that will affect the conductivity value. Also, some resistive materials are non-ohmic, meaning the current is not a linear function of the voltage. This nonlinearity is most often noticed at high voltage levels. For semiconductors, carriers may be generated by light or by high-frequency radiation. This excess carrier generation can be avoided by testing semiconductors in an adequately shielded dark chamber.

SURFACE MEASUREMENTS

Principles of the Method

The two most common approaches for measuring sheet or surface conductivity are the four-point probe method and the Van der Pauw method. Both approaches are similar to four-point measurements in that current is driven between a pair of probes or connections and the voltage is measured across the other two. The four-point probe method is most often realized by contacting a flat film surface with four equally spaced in-line probes (Valdes, 1954; Hall, 1967). The Van der Pauw method can measure resistivity of small, arbitrarily shaped layers where the four contacts are typically placed around the periphery of the sample.

The four-point probe method, as indicated in Figure 5, has four equally spaced in-line probes with probe tip diameters small compared to the probe spacing, s. An ohmic contact is assumed between the probe tip and the sample. Current is most commonly passed between the outer two probes, and the voltage difference is measured between the two inner probes. Resistivity in a four-point probe measurement is given by

$$\rho = 2\pi s F V/I \tag{5}$$

where F is a correction factor. For placement of probes near the center of a medium of area large relative to the probe spacing, and of a semi-infinite thickness, the correction factor, F, is unity. However, few practical measurements are made on semi-infinitely thick media. The factor F accounts for more realistic thickness and other nonidealities (Combs and Albert, 1963; Albert and Combs, 1964; Albers and Berkowitz, 1985). This factor is itself the product of a factor to correct for sample thickness, F_1, and a factor to account for lateral dimensions of the wafer, F_2. For a sample of conductive layer thickness, t, with a conducting bottom side

$$F_1 = \frac{t/s}{2\ln(\cosh(t/s)/\cosh(t/2s))} \tag{6}$$

If the bottom side is nonconducting,

$$F_1 = \frac{t/s}{2\ln(\sinh(t/s)/\sinh(t/2s))} \tag{7}$$

For a circular wafer of diameter, d, the correction factor F_2 is

$$F_2 = \frac{\ln 2}{\ln 2 + \ln\left[\dfrac{(d/s)^2+3}{(d/s)^2-3}\right]} \tag{8}$$

Note that F_2 approaches unity for $d > 40s$.

Locating the probes closer than four probe spacings from the wafer edge can also result in measurement error. Correction factors to account for edge proximity can be found in Schroder (1990).

The Van der Pauw method can determine the resistivity of small, arbitrarily shaped layers and generally requires less surface area than the four-point probe method (Van der Pauw, 1958). It is often used in integrated circuit processing. The method considers four small contacts placed around the periphery of a homogeneous, uniform thickness (t) sample, as indicated in Figure 6. In this figure, a resistance $R_{ab,cd}$ is determined by driving a current from point a to b and measuring the voltage from point c to d, or

$$R_{ab,cd} = \frac{|V_c - V_d|}{|I_{ab}|} \tag{9}$$

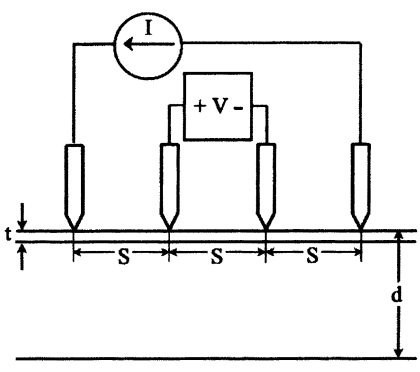


Figure 5. In-line four-point probe measurement of a conductive film of thickness, t, uses a known current source, high-impedance voltmeter, and spring-loaded sharp probes.

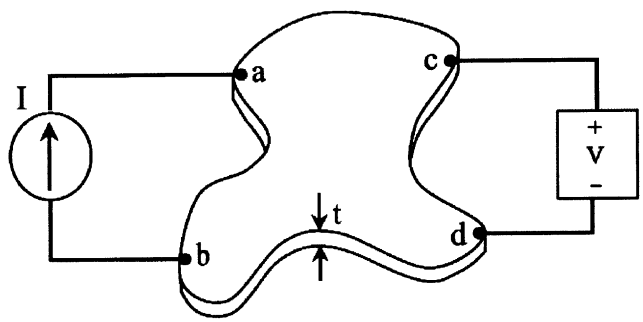


Figure 6. Van der Pauw measurement of an arbitrarily shaped sample uses a known current and a high-impedance voltmeter.

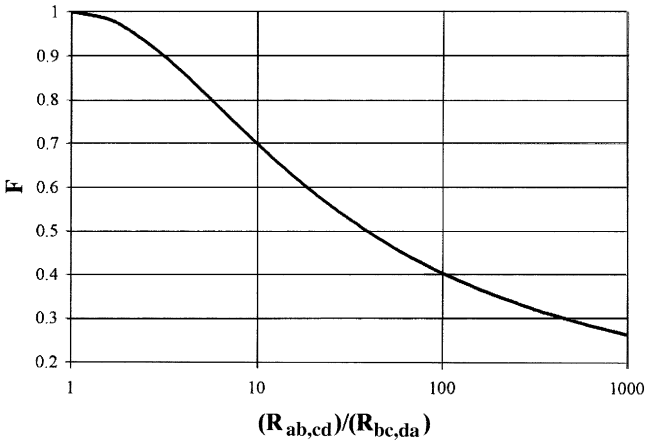


Figure 7. Van der Pauw function, F, plotted versus resistance ratio.

Using a conformal mapping approach Van der Pauw shows that

$$\exp\left[\frac{-\pi R_{\mathrm{ab,cd}} t}{\rho}\right] + \exp\left[\frac{-\pi R_{\mathrm{bc,da}} t}{\rho}\right] = 1 \qquad (10)$$

Solving for resistivity gives

$$\rho = \frac{\pi t}{\ln 2} \frac{R_{\mathrm{ab,cd}} + R_{\mathrm{bc,da}}}{2} F \qquad (11)$$

where the value of F is given by

$$\frac{R_{\mathrm{ab,cd}} - R_{\mathrm{bc,da}}}{R_{\mathrm{ab,cd}} + R_{\mathrm{bc,da}}} = \frac{F}{\ln 2} \operatorname{arccosh}\left[\frac{\exp(\ln 2/F)}{2}\right] \qquad (12)$$

This equation is used to plot F as a function of the resistance ratio $R_{\mathrm{ab,cd}}/R_{\mathrm{bc,da}}$ in Figure 7.

Equation 11 is simplified for 4-fold symmetrical shapes. For instance, the rotation of the structures in Figure 8 by 90° should result in the same measured resistance for a uniform-thickness, homogeneous film with identical contacts. In such a case, $F = 1$ and

$$\rho = \frac{\pi t}{\ln 2} R_{\mathrm{ab,cd}} = 4.532 t R_{\mathrm{ab,cd}} \qquad (13)$$

Practical Aspects of the Method

Like the four-point method used for bulk resistivity measurements, separating the current source from the high-impedance voltage meter avoids errors associated with contact resistance. Also, when considering semiconductor measurements, sufficient separation between the current and voltage probes is required so that minority carriers injected near the current probes recombine before their presence can be felt at the voltage probes. The probes

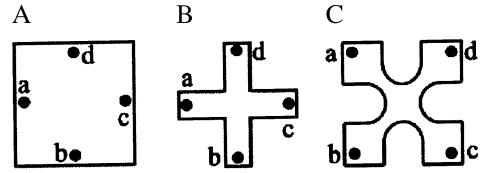


Figure 8. Common Van der Pauw structures: (**A**) square (contacts placed on edges, or placed on each corner), (**B**) "Greek Cross," and (**C**) clover leaf.

themselves have a tightly controlled spacing, typically 1 mm ± 0.01 mm, and the probe tips are individually spring-loaded with a force typically between 1 and 2 N to minimize damage to the film surface during probing. Hard tungsten probe tips are common, but osmium tips are also available, which can make good contact when heated on a semiconductor. A typical measurement range for such a system is 0.001 Ω-cm to 1000 Ω-cm.

In Van der Pauw measurements, it is common to calculate resistivity from two sets of measurements ($R_{\mathrm{ab,cd}}$ and $R_{\mathrm{bc,da}}$). For uniform samples with good contacts, the same results should be measured. The square pattern shown in Figure 8A, or a circular pattern with four points equidistant about the periphery, can be made very small for integrated-circuit film measurements and are often used in practice. However, because of probe alignment difficulties and problems making ideal probe to sample contacts, the "Greek Cross" of Figure 8B or the clover leaf structure of Figure 8C is often used. These shapes isolate the contacts and reduce the error encountered for finite-sized probe tips that are not located all the way out to the periphery of the sample.

Method Automation

As with bulk measurements, four-point probes are routinely connected to current sources and voltage meters for automated measurements. Four-point probing systems typically contain an adjustable constant current source used in conjunction with a high-impedance voltmeter. These systems automatically test and average both forward and reverse signals to reduce errors from thermal effects and rectifying contacts. More elaborate Van der Pauw measurement apparatus will make multiple measurements ($R_{\mathrm{ab,cd}}$ and $R_{\mathrm{bc,da}}$) and average the results.

Problems

The measured surface resistance depends on the doping profile, which affects both carrier concentration and carrier mobility. The conductivity of such a doped semiconductor is therefore a function of depth into the semiconductor. Measuring the surface resistance of a doped semiconductor assumes that the substrate below the junction depth is much more resistive than the layer to be measured. If the substrate is appreciably conductive, then a surface resistance measurement can still be made by forming a reverse-biased diode with the substrate.

NON-CONTACT METHODS

Principles of the Method

Contact-free test methods are useful when electrical contacts are difficult to make or when the sample must not be damaged. Also, a contactless measurement can be much quicker, since the time it takes to make contacts is eliminated. Eddy current approaches and relaxation techniques are often used for non-contact measurement. While such methods are not as accurate as the contact approaches discussed above, errors below 5% are achievable.

The most widely used approach for contactless measurement of conductivity is the eddy current method (ASTM E1004, 1991). In its most straightforward realization, an alternating current-carrying coil is placed close to a conductive sample. The magnetic fields generated by the coil induce circulatory currents, called eddy currents, in the sample which act to oppose the applied magnetic field. Small impedance changes are then measured in the coil as it is loaded by the conductive sample. The measured impedance is a function of sample conductivity, and also of the sample's mechanical structure near the surface. In fact, eddy current testing is routinely used to inspect the mechanical condition of conductive surfaces (Cartz, 1995).

Figure 9 illustrates the general measurement approach. The coil is driven by a sinusoidal current source, and measurement with a high-impedance voltmeter determines the coil impedance, $Z = V/I$. The current through the coil induces eddy currents in the conductive sample which act to oppose the change in magnetic flux. The induced eddy currents are in a plane parallel to a plane containing the coil loops and are a function of sample conductivity, magnetic permeability, and thickness, along with driving coil properties such as number of turns and distance from the sample. The coil itself may also be wrapped around or contained within a ferrite block to enhance the magnetic field.

An equivalent circuit for the simple test setup is shown in Figure 10, where the mutual inductance term M accounts for field coupling into the sample, which appears as a series RL circuit.

It can be shown (Cartz, 1995) that the impedance looking into the circuit is

$$Z = R_T + jX_T = \left[R + \frac{\omega^2 M^2 R'}{R'^2 + \omega^2 L'^2}\right] + j\left[\omega L - \frac{\omega^3 M^2 L'}{R'^2 + \omega^2 L'^2}\right] \quad (14)$$

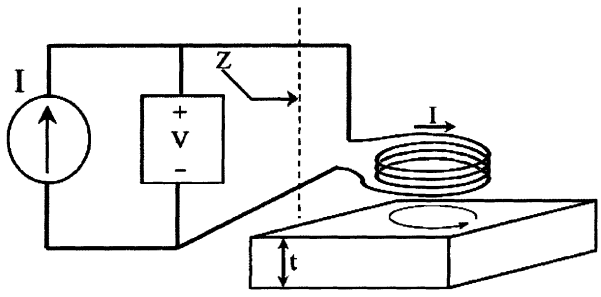


Figure 9. Single coil eddy-current measurement circuit. The impedance, Z, looking into the coil is changed by eddy currents induced in the sample.

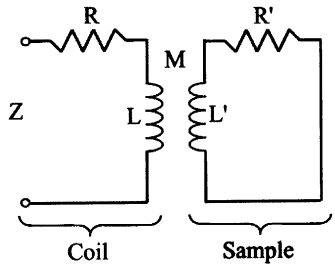


Figure 10. Equivalent circuit for the single coil eddy-current measurement approach.

where R_T and X_T represent the total real and imaginary parts of the impedance, respectively, and j is the imaginary square root of (-1). A plot of X_T versus R_T for a series of samples traces a conductivity curve as shown in Figure 11. This is a comparison method where samples of known conductivity are plotted, and the test sample behavior is then inferred from its location on the chart.

Another non-contact method is the relaxation approach, where a voltage step function applied to a resistive-reactive circuit will result in an exponential response. For instance, when the switch in Figure 12 is opened, the capacitor discharges through the resistor such that

$$V_r(\tau) = V_s \exp(-\tau/RC) \quad (15)$$

When time τ equals one "RC time constant," the voltage has dropped to e^{-1} of its initial value. Most often, this approach is used with a known value of resistance in order to measure the reactive component, but knowing the reactive component allows computation of the resistance. This is illustrated in Figure 13, where the voltage V_R from Figure 12 is initially charged to 1 V, and at time $\tau = 0$ the switch is thrown which allows this voltage to discharge through a 1-μF capacitor. The relaxation for three different resistance values is shown.

This relaxation approach has been used in conjunction with eddy currents where the sample is excited and the fields are measured as they decay (Kos and Fickett, 1994).

Practical Aspects of the Method

The eddy current approach requires two or more calibration standards covering the range of expected sample

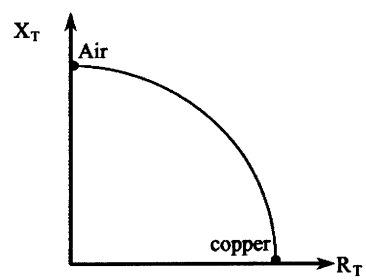


Figure 11. Impedance plane chart showing a typical conductivity curve.

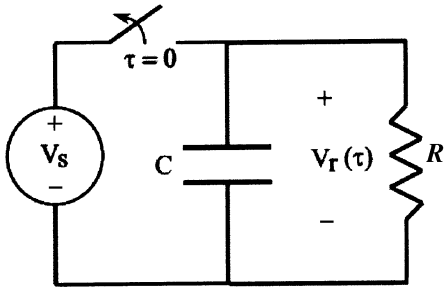


Figure 12. RC network used in relaxation method.

conductivity. Test conditions, such as driving current and frequency, must be the same for both calibration and test samples. The sample thickness and the distance between the coil and the sample surface (termed the "lift-off") must be carefully controlled. Finally, the coil should not be placed within two coil diameters of a sample discontinuity (such as a hole or edge).

In the simple method presented, the driving coil is also used for measurement. Improved eddy-current probes separate the driving and measuring coils, using them to sandwich the test sample. Since very little current is in the measurement coil, eddy currents induced in the sample are essentially all from the driving coil. The measurement coil sees fields from both the driving coil and from the eddy currents in the sample. For best sensitivity, the driving coil fields can be removed from the measurement using a lock-in amplifier circuit (Crowley and Rabson, 1976). Further improvement in the measurement circuit uses an additional set of coils for compensation. Such an approach has been used with miniature radio frequency (RF) coils to provide resistivity mapping with a 4 mm spatial resolution (Chen, 1989). An interesting variation on this method uses a driving coil and a pair of resonant coils to monitor in situ aluminum thin films deposited by chemical vapor deposition (Ermakov and Hinch, 1997).

The magnitude of the eddy currents attenuate by $e^{-y/\delta}$ as they penetrate a depth y into the sample, reaching a value of e^{-1} of the surface value at one skin depth, δ, given by

$$\delta = 1/\sqrt{\pi f \mu \sigma} \tag{16}$$

where f is frequency and μ is sample magnetic permeability. For a single-coil system, the effect of thickness is negligible for samples that are at least 2.6 skin depths thick. Note that for measuring surface defects, or the resistivity near the surface, higher frequencies may be employed. For a system where the measuring coil is on the other side of the sample from the driving coil, samples thinner than a skin depth are required.

The eddy current method is also used to measure conductivity of cylindrical samples. This is done by comparing the impedance of the coil with and without a cylindrical sample inserted (Zimmerman, 1961). Other approaches have relied on mutual inductance measured between a driving coil and measurement coil wrapped around the cylindrical sample (Rosenthal and Maxfield, 1975).

In the relaxation approach, the step function excites all possible resonances, with the one showing the longest decay time being proportional to the sample resistivity. It may be necessary to determine this time constant with a curve-fitting routine after sufficient time has passed for the other resonances to subside. Also, the time constant for the measurement coils must be much shorter than the sample time constant. This approach is most often used for low-resistivity samples (metals with $\rho < 10^{-4}$ ohm-micrometer), since otherwise the time constant is excessive.

A relaxation approach can also find semiconductor resistivity, provided the electrical permittivity is known. Here, a semiconductor wafer is inserted between conductive plates to form a parallel resistive-capacitive circuit. A step voltage is applied to the capacitors, and the current is measured versus time to arrive at an RC time constant equal to the product of the sample resistivity ρ and electrical permittivity, ε, called the "space charge relaxation time." Details of such a procedure used to map the resistivity profile across a gallium arsenide wafer are provided in Stibal et al. (1991).

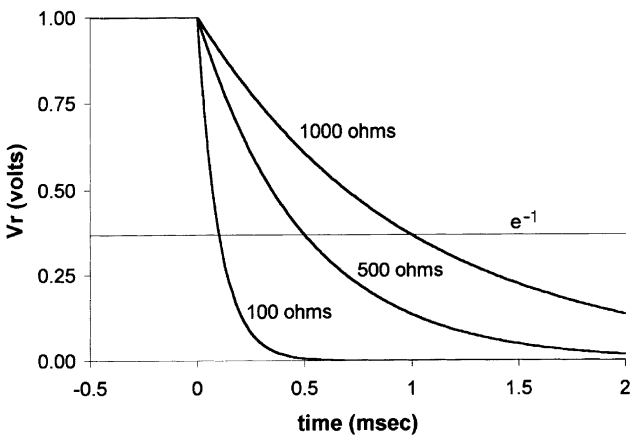


Figure 13. Response of the RC network with a 1-μF capacitor that has been initially charged to 1 V. The relaxation is shown for three values of resistance. The line at 0.368 indicates when the voltage has decayed to e^{-1} of its initial value.

MICROWAVE TECHNIQUES

Principles of the Method

A number of approaches exist to determine dielectric sample conductivity at high frequencies. The primary approaches use open-ended coaxial probes, coaxial or waveguide transmission lines, free-space radiation, and cavity resonators. Scattering parameter measurements are made using vector network analyzers, which are available at considerable expense, to perform measurements up to 10^{11} Hz. These scattering parameters can be related to the sample's electrical permittivity and loss tangent, from which the conductivity can be extracted.

From Maxwell's equations for time harmonic fields

$$\nabla \times \mathbf{H} = (\sigma + j\omega\varepsilon)\mathbf{E} \tag{17}$$

where E and H are the electric and magnetic fields, respectively, and ω is the angular frequency. Here we consider the conductivity σ and the permittivity ε within the dielectric. It is convenient to express Equation 17 as

$$\nabla \times \mathbf{H} = j\omega\varepsilon_c \mathbf{E} \tag{18}$$

where ε_c is a complex permittivity

$$\varepsilon_c = \varepsilon - j\frac{\sigma}{\omega} \tag{19}$$

Plotting the imaginary part of Equation 19 versus the real part reveals where the term loss tangent, tan δ, arises

$$\tan\delta = \frac{\sigma}{\omega\epsilon} \tag{20}$$

Here δ is the angle made with the real axis. Note that this δ has nothing to do with skin depth, which inconveniently shares the same symbol. The term loss tangent is typically applied when discussing dielectric materials, for which a small value is desirable.

Extracting the complex permittivity from scattering parameter data is far from trivial and is almost always performed using software within the network analyzer or in post-processing. In the cavity resonator, waveguide transmission line, and free space approaches, it is also possible to extract the material's complex permeability from the measured scattering parameters (Application Note 1217; see Literature Cited). This can be useful for the study of magnetic materials.

Practical Aspects of the Method

The four general approaches are briefly described below, along with suitable references. Of particular interest are the publications of Bussey (1967) and Baker-Jarvis et al. (1992).

Coaxial probes. In this approach, the open end of a coaxial probe is placed against the flat face of a solid sample as shown in Figure 14A. The probe may also be immersed in a liquid sample. The fringing fields between the center and the outer conductors are affected by the sample material. Complex permittivity is extracted from measurements of the reflection scattering parameter S_{11}. This approach is typically used to measure complex permittivity over a frequency range from 2×10^8 Hz up to 2×10^{10} Hz, and has the advantage that no special sample preparation is required. In addition, stainless steel coaxial probes have performed at temperatures up to 1000°C (Gershon et al., 1999).

Transmission lines. In the various transmission line methods, a material sample is snugly inserted into a section of coaxial transmission line, rectangular waveguide, or circular waveguide (Srivastava and Jain, 1971; Ligthart, 1983; Wolfson and Wentworth, 2000). More than one measurement is required for extraction of complex permittivity data from one port measurements. This can be accomplished by using more than one length of sample or by terminating the guide containing the sample in more than one load. The measured complex reflection coefficients can be manipulated to extract complex permittivity. Two-port measurements can be performed on a single sample where in addition to reflection, the transmission properties are measured. Precise machining of the samples to fit inside the transmission line is critical, which can be somewhat of a disadvantage for the coaxial approach (shown in Figure 14B) where an annular sample is required. An advantage of the coaxial approach is that measurements can be made over a much broader frequency range than is possible with the waveguide procedure. Common problems for the transmission line techniques include air gaps, resonances at sample lengths corresponding to half wavelengths, and excitation of higher-order propagating modes.

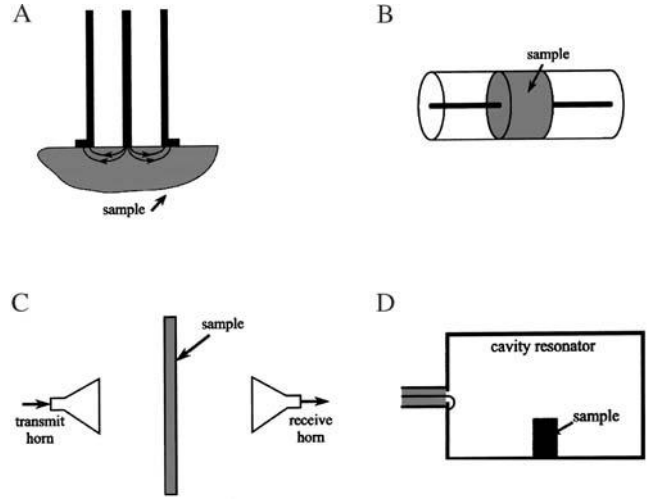


Figure 14. Microwave measurement approaches: (**A**) open-ended coaxial probe in contact with sample, (**B**) sample inserted into coaxial transmission line, (**C**) free-space approach where sample is placed between horn antennas, and (**D**) sample placed within a cavity resonator.

Free-space radiation. Microwaves are passed through material samples using transmitting and receiving antennas as shown in Figure 14C. In addition to measurements through the sample, the transmitting and receiving antennas may be placed at angles on the same side of the sample and a reflection measurement taken. The samples used in free-space techniques must be fairly large, with flat surfaces. Measurement approaches and data processing proceeds as with the transmission line approaches. The free-space approach is a noncontacting method that can be especially useful for making measurements at high temperature or under harsh environmental conditions. It has been employed in measurements ranging from 2×10^9 Hz up to 1.1×10^{11} Hz (Otsuka et al., 1999).

Cavity resonators. Here, a sample of precisely known geometry is placed within a microwave cavity and changes in the resonant frequency and the resonator Q are measured and processed to yield the complex permittivity (Kraszewski and Nelson, 1992). As shown in Figure 14D,

a coaxial probe inserted into the center of one wall of the cavity is used to supply energy for resonance. If the sample dimensions are precisely known, and if calibration standards of the same dimensions are available, then this approach can yield very accurate results for both complex permittivity and complex permeability, although it tends to have limited accuracy for low-loss materials. It has been employed in measurements ranging from 5×10^8 Hz up to 1.1×10^{11} Hz.

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